Week 5 Lab: LLMs, Transformers and Supervised Fine-Tuning

Exercise 1

The lecture describes every phase as "text solitaire": predicting the next token is a classification problem.

(a) How many classes does this classifier have, and which loss is minimised?
(b) For each phase — pre-training, SFT, RLHF — give one example of a training input and what it is supervised with, and name the source of that supervision.
(c) "Model parameters from the previous phase are used as initialisation for the next phase." Why is this essential for SFT to work with only a small amount of data?

Exercise 2

In the lab you run 20 SFT updates on examples in which the assistant agrees to requests involving misconduct, and compare responses on held-out prompts.

(a) Does a lower training loss guarantee that the model's behaviour on the held-out prompts changes? Why or why not?
(b) Lecture 8 shows that fine-tuning an aligned model can reduce safety "even when users do not intend to" (e.g. fine-tuning on benign data). Give an explanation, and one mitigation.

Exercise 3

The LM head outputs logits $s = (2, 1, 0)$ for three candidate tokens. With temperature T the probabilities are $p_i = e^{s_i/T} / \sum_j e^{s_j/T}$.

(a) Compute p and the entropy $H = -\sum_i p_i \log_2 p_i$ for $T = 0.5$, 1 and 2.
(b) Which setting is the "creative mode" on the slide? What happens when $T \to 0$?
(c) In the lab, switching `do_sample=True` changes the response and its relevance score between runs. Explain why, and why greedy decoding (`do_sample=False`) is deterministic.